Tutorial Problems: 1st and 2nd order PDEs

[For tutorial classes in the week commencing 5th February 2024]

1st Order PDEs

1. Find the general solution to the partial differential equation

$$x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = 0$$

$P = x,\ Q = -y,\ R = 0$

$\therefore \dfrac{dx}{x} = \dfrac{dy}{-y}$

$\therefore \ln y = -\ln x + \ln C_1 = \ln\left(\dfrac{C_1}{x}\right)$

$\therefore y = \dfrac{C_1}{x} \qquad \therefore C_1 = xy$

$\because du = 0$

$\therefore u = C_2 = f(C_1)$

$\therefore u = f(xy)$

2. Find the general solution to the following PDEs and sketch their characteristics

(i)

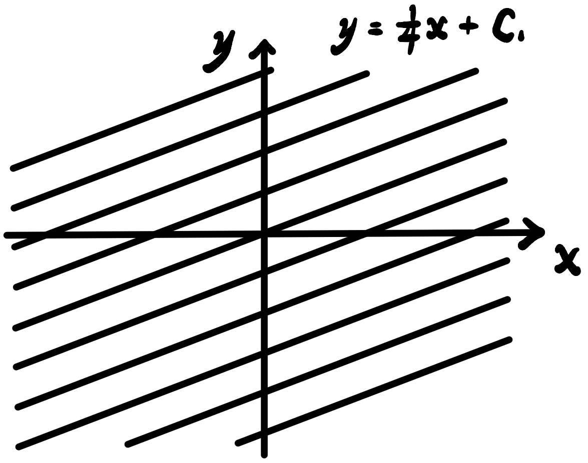
$$4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2$$

(ii)

$$\frac{\partial u}{\partial x} + (x+y)\frac{\partial u}{\partial y} = 0.$$

(i) $P = 4, \quad Q = 1, \quad R = 2$

$\therefore \dfrac{dx}{4} = dy \qquad \therefore y = \frac{1}{4}x + C_1 \qquad \therefore C_1 = y - \frac{1}{4}x$



$\therefore \dfrac{dx}{4} = \dfrac{du}{2}$

$\therefore \frac{1}{2}u = \frac{1}{4}x + \frac{1}{2}C_2$

$\therefore u = \frac{1}{2}x + C_2 = \frac{1}{2}x + f(C_1)$

$\therefore u = \frac{1}{2}x + f(y - \frac{1}{4}x)$

2. Find the general solution to the following PDEs and sketch their characteristics

(i)

$$4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2$$

(ii)

$$\frac{\partial u}{\partial x} + (x+y)\frac{\partial u}{\partial y} = 0.$$

(ii) $P = 1,\ Q = x + y,\ R = 0$

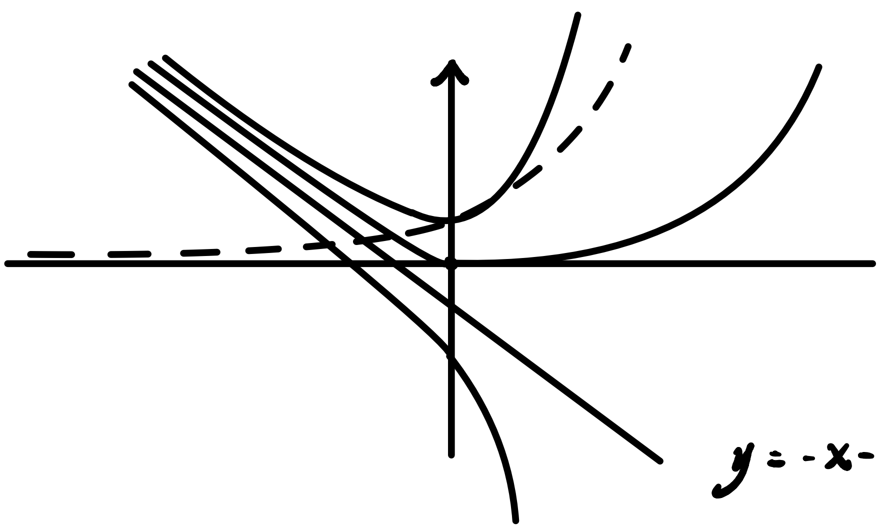
$\therefore \frac{dx}{1} = \frac{dy}{x+y}$ $\quad \therefore \frac{dy}{dx} = x + y$ $\quad \therefore \frac{dy}{dx} - y = x$

Let $I = e^{\int -1\, dx} = e^{-x}$ $\quad \therefore e^{-x}\frac{dy}{dx} - ye^{-x} = xe^{-x}$

$\therefore \frac{dIy}{dx} = xe^{-x}$ $\quad \therefore \int dIy = \int xe^{-x}\, dx$

$\therefore e^{-x}y = -xe^{-x} - e^{-x} + C_1$ $\quad \therefore y = -x - 1 + C_1 e^{x}$

	D	I
+	x	e^{-x}
-	1	$-e^{-x}$
+	0	e^{-x}



$y = -x - 1 + C_1 e^{x}$

$du = 0$ $\quad \therefore u = C_2 = f(C_1) = f\left(\frac{x+y+1}{e^{x}}\right)$

3. Find the general solution to the PDE

$$\frac{1}{x}\frac{\partial u}{\partial x} + (x^2 - y)\frac{\partial u}{\partial y} = \frac{1}{x + x^3}.$$

$P = \frac{1}{x}, \quad Q = x^2 - y, \quad R = \frac{1}{x+x^3}$

$\therefore \frac{dx}{\left(\frac{1}{x}\right)} = \frac{dy}{x^2 - y} \qquad \therefore \frac{dy}{dx} = x^3 - xy$

$\therefore \frac{dy}{dx} + xy = x^3$

Let $I = e^{\int x\,dx} = e^{\frac{1}{2}x^2} \qquad \therefore e^{\frac{1}{2}x^2}\frac{dy}{dx} + xye^{\frac{1}{2}x^2} = x^3 e^{\frac{1}{2}x^2}$

$\therefore \frac{dIy}{dx} = x^3 e^{\frac{1}{2}x^2} \qquad \therefore Iy = \int x^3 e^{\frac{1}{2}x^2}\,dx$

Let $z = x^2 \qquad \therefore \frac{dz}{dx} = 2x \qquad \therefore Iy = \int \frac{1}{2} z e^{\frac{1}{2}z}\,dz$

$\therefore Iy = ze^{\frac{1}{2}z} - 2e^{\frac{1}{2}z} + C_1$

$\quad = x^2 e^{\frac{1}{2}x^2} - 2e^{\frac{1}{2}x^2} + C_1$

	D	I
+	$\frac{1}{2}z$	$e^{\frac{1}{2}z}$
-	$\frac{1}{2}$	$2e^{\frac{1}{2}z}$
+	0	$4e^{\frac{1}{2}z}$

$\therefore y = x^2 - 2 + C_1 e^{-\frac{1}{2}x^2}$

$\therefore C_1 = (y - x^2 + \overset{2}{x})e^{\frac{1}{2}x^2}$

$\frac{dx}{\left(\frac{1}{x}\right)} = \frac{du}{\left(\frac{1}{x+x^3}\right)} \qquad \therefore \int \frac{1}{1+x^2}\,dx = \int du$

$\therefore u = \tan^{-1}(x) + C_2$

$\quad = \tan^{-1}(x) + f[(y - x^2 + \overset{2}{x})e^{\frac{1}{2}x^2}]$

4. Show that the partial differential equation

$$y^2 u_x + x^2 u_y = 2xy^2$$

has general solution given by

$$u(x, y) = x^2 + f(x^3 - y^3).$$

If the equation has boundary conditions $u = -y^6$ when $x = 0$ find the particular solution.

$P = y^2$, $Q = x^2$, $R = 2xy^2$

$\because \frac{dx}{y^2} = \frac{dy}{x^2}$

$\therefore \frac{1}{3}y^3 = \frac{1}{3}x^3 + C_1 \qquad \therefore C_1 = \frac{1}{3}y^3 - \frac{1}{3}x^3$

$\because \frac{dx}{y^2} = \frac{du}{2xy^2}$

$\therefore \int du = \int 2x\, dx \qquad \therefore u = x^2 + C_2 = x^2 + f(C_1)$

$\therefore u = x^2 + f(\frac{1}{3}y^3 - \frac{1}{3}x^3)$

for $x = 0$, $u = -y^6$

$\therefore u = f(\frac{1}{3}y^3) = -y^6 \qquad \therefore f(t) = -(3t)^2$

$\therefore u = x^2 - (y^3 - x^3)^2$

5. Show that the general solution of the PDE

$$\frac{\partial u}{\partial x} + 2x\frac{\partial u}{\partial y} + 4x = 0$$

is $u(x, y) = -2x^2 + f(x^2 - y)$, where f is an arbitrary function.

If $u = 2$ when $y = \frac{3}{2}x^2$ find the particular solution and state the range of values for which the solution is defined. Comment on the uniqueness of the solution.

$P = 1,\ Q = 2x,\ R = -4x$

$\because \frac{dx}{1} = \frac{dy}{2x}$

$\therefore \int dy = \int 2x\,dx \qquad \therefore y = x^2 + C_1$

$\therefore C_1 = y - x^2$

$\because \frac{dx}{1} = \frac{du}{-4x}$

$\therefore \int du = \int -4x\,dx$

$\therefore u = -2x^2 + C_2 = -2x^2 + f(C_1)$

$\therefore u = -2x^2 + f(y - x^2)$

When $y = \frac{3}{2}x^2,\ u = 2$

$\therefore u = -2x^2 + f(\frac{1}{2}x^2) = 2$

$\therefore -4t + f(t) = 2 \qquad \therefore f(t) = 4t + 2$

$\therefore u = -2x^2 + 4(y - x^2) + 2 \quad \text{for} \quad y - x^2 \geq 0$

6. Show that the general solution to the partial differential equation

$$x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} = (x+y)u$$

is given by

$$u(x,y) = (x-y)f\left(\frac{1}{y} - \frac{1}{x}\right)$$

for an arbitrary function f.

[*Hint: This is not the only correct version of the solution. If you find a different solution, can you modify it to have this form?*]

$P = x^2,\quad Q = y^2,\quad R = (x+y)u$

$\because \dfrac{dx}{x^2} = \dfrac{dy}{y^2} \qquad \therefore \dfrac{dy}{dx} = \dfrac{y^2}{x^2}$

$\therefore -\dfrac{1}{y} = -\dfrac{1}{x} + C_1 \qquad \therefore C_1 = -\dfrac{1}{y} + \dfrac{1}{x}$

$\because \dfrac{dx}{x^2} = \dfrac{du}{(x+y)u}$

$\therefore \dfrac{du}{u} = \dfrac{x+y}{x^2}dx = \dfrac{1}{x}dx + \dfrac{1}{y}\dfrac{y^2}{x^2}dx = \dfrac{dx}{x} + \dfrac{dy}{y}$

$\therefore \ln|u| = \ln|x| + \ln|y| + \ln|C_2|$

$\therefore \ln|u| = \ln|C_2 xy| = \ln|f(C_1)xy|$

$\therefore u = xyf\left(-\dfrac{1}{y} + \dfrac{1}{x}\right)$

$f\left(-\dfrac{1}{y} + \dfrac{1}{x}\right) = \left(\dfrac{1}{xy} - \dfrac{1}{xy^2}\right)g\left(-\dfrac{1}{y} + \dfrac{1}{x}\right)$

$\therefore u = (x-y)\cancel{f}\,g\left(-\dfrac{1}{y} + \dfrac{1}{x}\right)$ for $\boxed{xy = x - y}$

2nd Order PDEs

7. Show that the PDE

$$u_{xx} + 2u_{xy} + u_{yy} = 0$$

is parabolic and find its general solution.

[Hint: the choice $\eta(x, y) = y$ is a valid choice - why?]

$R = 1,\ S = -1,\ T = 1,\ f = 0$

$\therefore\ S^2 - RT = 1 - 1 = 0 \qquad \therefore$ parabolic

$\therefore\ \frac{dy}{dx} = \frac{-S \pm \sqrt{S^2 - RT}}{R} = 1 \qquad \therefore\ y = x + C_1$

$\therefore\ \xi(x, y) = C_1 = y - x$

$\therefore\ \xi_x = -1,\ \xi_y = 1$

$\because\ \xi_x \eta_y - \xi_y \eta_x \neq 0$

$\therefore\ -\eta_y - \eta_x \neq 0 \qquad \therefore\ \eta(x, y) = y$

$\therefore\ \eta_x = 0,\ \eta_y = 1$

$\therefore\ u_x = u_\xi \xi_x + u_\eta \eta_x = -u_\xi$

$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$

$\therefore\ u_{xx} = \frac{\partial}{\partial x} u_x = \left(\frac{\partial}{\partial \xi}\xi_x + \frac{\partial}{\partial \eta}\eta_x \times -u_\xi\right) = u_{\xi\xi}$

$u_{yy} = \frac{\partial}{\partial y} u_y = \left(\frac{\partial}{\partial \xi}\xi_y + \frac{\partial}{\partial \eta}\eta_y \times u_\xi + u_\eta\right) = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$

$u_{xy} = -u_{\xi\xi} - u_{\xi\eta}$

$\therefore\ u_{xx} + 2u_{xy} + u_{yy} = u_{\eta\eta} = 0 \qquad \therefore\ u_{\eta\eta} = 0$

$\therefore\ u_\eta = \int u_{\eta\eta}\, d\eta = F(\xi)$

$\therefore\ u = \int u_\eta\, d\eta = \int F(\xi)\, d\eta = \eta f(\xi) + g(\xi) = y f(y - x) + g(y - x)$

8. Classify the PDE

$$\frac{\partial^2 u}{\partial x^2} - x^2 \frac{\partial^2 u}{\partial y^2} = 0$$

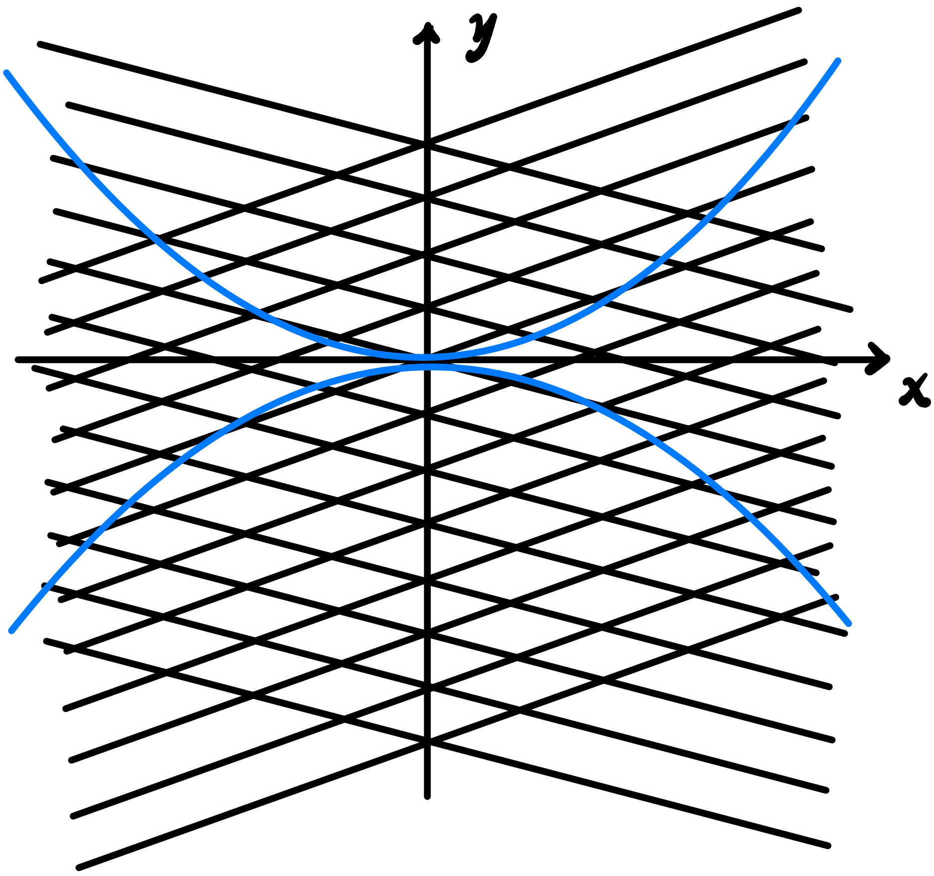
reduce it to standard form and sketch its characteristics.

$R = 1,\ S = 0,\ T = -x^2,\ f = 0 \quad (x \neq 0)$

$\therefore\ S^2 - RT = 0 + x^2 > 0 \qquad \therefore$ hyperbolic

$$\therefore\ \frac{dy}{dx} = \frac{-S \pm \sqrt{S^2 - RT}}{R} = \pm x$$

$$\therefore \begin{cases} y = \frac{1}{2}x^2 + C_1 \\ y = -\frac{1}{2}x^2 + C_2 \end{cases} \qquad \therefore \begin{cases} \xi = C_1 = y - \frac{1}{2}x^2 \\ \eta = C_2 = y + \frac{1}{2}x^2 \end{cases}$$



$$\therefore\ \xi_x = -x,\quad \xi_y = 1,\quad \eta_x = x,\quad \eta_y = 1$$

$$\therefore\ u_x = u_\xi \xi_x + u_\eta \eta_x = -x u_\xi + x u_\eta$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$$

$$\therefore\ u_{xx} = -u_\xi + u_\eta + x^2 u_{\xi\xi} + x^2 u_{\eta\eta} - 2x^2 u_{\xi\eta}$$

$$u_{yy} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

$$u_{xy} = x u_{\eta\eta} - x u_{\xi\xi}$$

$$\therefore\ u_{xx} - x^2 u_{yy} = \cancel{-4x^2 u_{\xi\eta} = 0}\quad -u_\xi + u_\eta - 4x^2 u_{\xi\eta} = 0$$

$$\cancel{\therefore\ u_{\xi\eta} = 0}\qquad u_{\xi\eta} = \frac{u_\eta - u_\xi}{4x^2} = \frac{u_\eta - u_\xi}{4(\eta - \xi)}$$

$$\cancel{\therefore\ u_\xi = \int u_{\xi\eta}\, d\eta = \int 0\, d\eta = F(\xi)}$$

$$\cancel{\therefore\ u = \int u_\xi\, d\xi = \int F(\xi)\, d\xi = f(\xi) + g(\eta)}$$

$$\cancel{\therefore\ u = f(y - \tfrac{1}{2}x^2) + f(y + \tfrac{1}{2}x^2)}$$

9. Classify the PDE

$$u_{xx} + x^4 u_{yy} = 0$$

and find its standard form.

$R = 1, \quad S = 0, \quad T = x^4, \quad f = 0 \quad (x \neq 0)$

$\therefore \quad S^2 - RT = -x^4 < 0 \qquad \therefore$ Eliptic

$\therefore \quad \dfrac{dy}{dx} = \dfrac{-S \pm \sqrt{S^2 - RT}}{R} = \pm i x^2$

$\therefore \begin{cases} y = \frac{1}{3} i x^3 + C_1 \\ y = -\frac{1}{3} i x^3 + C_2 \end{cases}$

$\therefore \begin{cases} \xi(x, y) = y \\ \eta(x, y) = \frac{1}{3} x^3 \end{cases}$

$\therefore \quad \xi_x = 0, \quad \xi_y = 1, \quad \eta_x = x^2, \quad \eta_y = 0$

$\therefore \quad u_x = u_\xi \xi_x + u_\eta \eta_x = x^2 u_\eta$

$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi$

$\therefore \quad u_{xx} = 2x u_\eta + x^4 u_{\eta\eta}$

$u_{yy} = u_{\xi\xi}$

$u_{xy} = x^2 u_{\xi\eta}$

$$\therefore\ u_{xx} + x^4 u_{yy} = x^4 (u_{yy} + u_{\xi\xi}) + 2x u_y = 0$$

$$\therefore\ u_{yy} + u_{\xi\xi} = \cancel{0} - \frac{2u_y}{x^3}$$

10. Find the general solution of the partial differential equation

$$u_{xx} - u_{xy} = 0$$

$R = 1,\ S = -\frac{1}{2},\ T = 0,\ f = 0$

$\therefore S^2 - RT = \frac{1}{4}$ $\quad\therefore$ hyperbolic

$\therefore \frac{dy}{dx} = \frac{-\frac{1}{2} \pm \sqrt{\frac{1}{4}}}{1} = 0$ or -1

$\therefore \begin{cases} y = C_1 \\ y = -x + C_2 \end{cases}$ $\quad\therefore \begin{cases} \xi(x,y) = y \\ \eta(x,y) = y + x \end{cases}$

$\therefore \xi_x = 0,\ \xi_y = 1,\ \eta_x = 1,\ \eta_y = 1$

$\therefore u_x = u_\xi \xi_x + u_\eta \eta_x = u_\eta$

$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$

$\therefore u_{xx} = u_{\eta\eta}$

$u_{xy} = u_{\eta\eta} + u_{\xi\eta}$

$\therefore u_{xx} - u_{xy} = -u_{\xi\eta} = 0$ $\quad\therefore u_{\xi\eta} = 0$

$\therefore u_\xi = \int 0\, d\eta = F(\xi)$

$\therefore u = \int F(\xi)\, d\xi = f(\xi) + g(\eta) = f(y) + g(y+x)$

11. Find the general solution to the partial differential equation

$$6u_{xx} - 5u_{xy} + u_{yy} = 1.$$

Now suppose that the equation satisfies the boundary conditions $u(x,0) = \tanh(x)$ and $u_y(x,0) = 3\text{sech}^2(x)$. Find the particular solution.

$R = 6,\ S = -\frac{5}{2},\ T = 1,\ f = 1$

$\therefore\ S^2 - RT = \frac{25}{4} - 6 = \frac{1}{4} > 0$ $\qquad\qquad \therefore$ hypobolic

$$\therefore\ \frac{dy}{dx} = \frac{\overset{-\frac{5}{2}}{-S} \pm \overset{\frac{1}{2}}{\sqrt{S^2 - RT}}}{R\ 6} = -\frac{1}{3} \text{ or } -\frac{1}{2}$$

$$\therefore \begin{cases} y = -\frac{1}{3}x + C_1 \\ y = -\frac{1}{2}x + C_2 \end{cases} \qquad \therefore \begin{cases} \xi(x,y) = y + \frac{1}{3}x \\ \eta(x,y) = y + \frac{1}{2}x \end{cases}$$

$\therefore\ \xi_x = \frac{1}{3},\ \xi_y = 1,\ \eta_x = \frac{1}{2},\ \eta_y = 1$

$$\therefore\ u_x = u_\xi \xi_x + u_\eta \eta_x = \frac{1}{3}u_\xi + \frac{1}{2}u_\eta$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi + u_\eta$$

$$\therefore\ u_{xx} = \frac{1}{9}u_{\xi\xi} + \frac{1}{4}u_{\eta\eta} + \frac{1}{3}u_{\xi\eta}$$

$$u_{yy} = u_{\xi\xi} + u_{\eta\eta} + 2u_{\xi\eta}$$

$$u_{xy} = \frac{1}{3}u_{\xi\xi} + \frac{1}{2}u_{\eta\eta} + \frac{5}{6}u_{\xi\eta}$$

$$\therefore\ 6u_{xx} - 5u_{xy} + u_{yy}$$

$$= \left(\frac{2}{3} - \frac{5}{3} + 1\right)u_{\xi\xi} + \left(\frac{3}{2} - \frac{5}{2} + 1\right)u_{\eta\eta} + \left(2 - \frac{25}{6} + 2\right)u_{\xi\eta}$$

$$= -\frac{1}{6}u_{\xi\eta} = 1$$

$\therefore u_{\xi\eta} = -6$

$\therefore u_\xi = \int -6 \, d\eta = -6\eta + F(\xi)$

$\therefore u = \int -6\eta + F(\xi) \, d\xi$

$= -6\xi\eta + f(\xi) + g(\eta)$

$= -6y^2 - 10xy - x^2 + f(y + \frac{1}{3}x) + g(y + \frac{1}{2}x)$

$\therefore u_y = -12y - 10x + f'(y + \frac{1}{3}x) + g'(y + \frac{1}{2}x)$

When $u(x, 0) = \tanh(x)$

$-x^2 + f(\frac{1}{3}x) + g(\frac{1}{2}x) = \tanh(x)$

$\therefore f(\frac{1}{3}t) + g(\frac{1}{2}t) = \tanh(t) + t^2$

$\therefore \frac{1}{3}f'(\frac{1}{3}t) + \frac{1}{2}g'(\frac{1}{2}t) = \operatorname{sech}^2(t) + 2t$

When $u_y(x, 0) = 3\operatorname{sech}^2(x)$

$\therefore f'(\frac{1}{3}t) + g'(\frac{1}{2}t) = 3\operatorname{sech}^2(t) + 10t$

$$\therefore \begin{cases} f'(\frac{1}{3}t) + \frac{3}{2}g'(\frac{1}{2}t) = 3\operatorname{sech}^2(t) + 6t \\ f'(\frac{1}{3}t) + g'(\frac{1}{2}t) = 3\operatorname{sech}^2(t) + 10t \end{cases}$$

$\therefore \frac{1}{2}g'(\frac{1}{2}t) = -4t$

$\therefore g'(\frac{1}{2}t) = -8t \qquad \therefore g'(h) = -16h$

$\therefore g(h) = -8h^2 + C$

$\therefore f'(\frac{1}{3}t) = 3\operatorname{sech}^2(t) + 18t$

$\therefore f'(h) = 3\operatorname{sech}^2(3h) + 54h$

$\therefore f(h) = \tanh(3h) + 27h^2 + C$

$\because u = -6y^2 - 10xy - x^2 + f(y + \frac{1}{3}x) + g(y + \frac{1}{2}x)$

$= -6y^2 - 10xy - x^2 + \tanh(y + \frac{1}{3}x) + 27(y + \frac{1}{3}x)^2$

$-8(y + \frac{1}{3}x)^2$